

Luiz Carazolli

M.Sc. candidate in Mathematics | University of Campinas

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Profile

I'm a CAPES-funded master's candidate in Mathematics at Unicamp, where I also earned my bachelor's degree. My research connects algebraic geometry and symplectic topology, with a focus on orbifold Gromov-Witten theory and quantum cohomology. Alongside this work, I'm developing interests in topological data analysis and the mathematical foundations of deep learning, building on earlier applied AI research. I'm seeking an entry-level industry research role in mathematics, finance, or AI, and welcome connections across geometry, topology, and data science.

Education

M.Sc. in Mathematics

2025-present | University of Campinas (Unicamp)

Research: Orbifold Quantum Cohomology.

Advisors: Prof. Dr. Lino Grama and Dr. Leonardo Cavenaghi. Funding: CAPES.

B.Sc. in Mathematics

2021-2024 | University of Campinas (Unicamp)

Research experience

Special Lagrangian Manifolds via Examples

2024 | International Center for Mathematical Sciences · Bulgarian Academy of Sciences

An introduction to special Lagrangian submanifolds through concrete examples in complex Euclidean space and Calabi-Yau threefolds.

Supervision: Prof. Dr. Ludmil Katzarkov and Prof. Dr. Lino Grama. Funding: FAPESP-BEPE · Grant 2024/07883-0.

Introduction to Symplectic Geometry

2023-2024 | University of Campinas

Foundations of symplectic geometry, with a study of coadjoint orbits of $SL(2, \mathbb{R})$ and an introduction to generalized complex geometry.

Supervision: Prof. Dr. Lino Grama. Funding: FAPESP · Grant 2023/07412-4.

Differential Forms and the Geometry of Surfaces

2022-2023 | University of Campinas

Differential forms and Cartan's moving-frame method, culminating in a proof of the Gauss-Bonnet theorem for surfaces.

Supervision: Prof. Dr. Lino Grama. Funding: CNPq.

Artificial Intelligence & Deep Learning

2022 | GoToData

Applied research experience in deep learning, natural language processing, and deep neural networks. Funding: MCTI.

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Curriculum Vitae | October 2026

Publications

A quantum connection approach to rationality

2026 | Preprint | arXiv:2610.06669

Luiz Carazolli, Leonardo F. Cavenaghi, Ludmil Katzarkov, Maxim Kontsevich, and Pedro Antonio Muniz Martins

<https://arxiv.org/abs/2610.06669>

Teaching

Teaching Assistant - General Topology

2026 | IMECC, University of Campinas

Graduate-level course.

Teaching Assistant - Introduction to Analysis

2025 | IMECC, University of Campinas

Undergraduate-level course.

Technical skills

Python · R · Mathematica · LaTeX

Academic and project-oriented research, mathematical writing, and teaching.

Selected talks

Introduction to Orbifolds

2024 | Seminar

The Gauss-Bonnet Theorem via the Moving-Frame Method

2023 | 34th Brazilian Mathematics Colloquium

Differential Forms and the Geometry of Surfaces

2023 | Unicamp Undergraduate Research Congress

Selected training

Foundations of Business Strategy - University of Virginia, 2025.

Differentiable Manifolds - UFMG, 2024.

Introduction to Banach Spaces - IMPA, 2023.

General Topology - UFMG, 2023.

Languages

Portuguese (native) · English (C2 proficiency, 2017)

Spanish and Italian (intermediate).

Selected honors

Gold medals: National Science Olympiad, Brazilian Astronomy Olympiad, and Unicamp Mathematics Olympiad (2020).

Silver medal: Brazilian Physics Olympiad (2020).